

NakedAgent: A Zero-Dependency Coding Agent Architecture via Standard-Library Primitives and Provable Replay

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ABSTRACT

Autonomous software engineering agents driven by Large Language Models (LLMs) increasingly rely on expansive, multi-tiered dependency trees. Mainstream agent scaffolds (e.g., SWE-agent, Aider, OpenHands) import between 20 and 50+ third-party packages to facilitate HTTP communication, terminal UI rendering, provider protocol abstraction, and telemetry. While feature-rich, this architectural paradigm introduces substantial supply-chain vulnerability attack surfaces, runtime version fragility, and heavy deployment footprints. In this paper, we introduce **NakedAgent**, an autonomous coding agent engineered under a strict zero-runtime-dependency constraint: it executes entirely upon standard Python 3.10+ library modules (`urllib`, `json`, `subprocess`, `re`, and `hashlib`). NakedAgent decouples tool invocation from proprietary JSON function-calling schemas by utilizing a Markdown fenced code-block protocol, enabling consistent tool use across local models (such as Qwen 2.5-Coder and Gemma 3) and hosted endpoints. To establish auditability and non-repudiation, we model agent execution as a pure Mealy state machine reducer $(S_t, E_t) \rightarrow (S_{t+1}, A_{t+1})$, combined with SHA-256 Merkle-linked state hashing recorded in an append-only JSON Lines (JSONL) event log. This architecture enables zero-inference offline replay verification, validating tool execution histories without querying an LLM. We present empirical findings from a SWE-bench Lite pilot with local 12B models, delineating scaffold-induced failure modes from model-inherent reasoning bounds. Finally, we formulate the *Omakase* extensibility doctrine, demonstrating how clean tool substitution seams empower extensible agent behaviors without framework bloat.

1. INTRODUCTION

Large Language Models have catalyzed a dramatic transformation in software development automation [1, 2]. Contemporary coding agents iteratively examine codebases, analyze compiler and test diagnostics, formulate code patches, and execute arbitrary shell commands within closed feedback loops [6, 7].

However, existing state-of-the-art coding agent frameworks—including Aider [3], SWE-agent [2], gptme [4], and OpenHands [5]—rely on extensive dependency graphs. Typical installations depend on dozens of transitive packages spanning asynchronous networking (`httpx`, `aiohttp`), rich terminal styling (`rich`, `prompt_toolkit`), telemetry aggregators, and proprietary provider SDKs.

This dependency proliferation incurs three primary systemic liabilities:

- 1. Supply-Chain Vulnerability Surfaces:** Every additional library introduced into an agent runtime expands its attack surface [10, 11]. Because software engineering agents routinely hold arbitrary execution and filesystem modification privileges within their execution environment, any vulnerability or compromise in a transitive dependency grants immediate arbitrary code execution.
- 2. Runtime Fragility and Upstream Drift:** Rapid version churning across transitive dependencies frequently causes breaking behavioral drifts, malformed serialization, or broken imports that destabilize autonomous workflows.

- 3. Deployment Footprint:** In air-gapped enclaves, high-security infrastructure, and resource-constrained edge machines, verifying and containerizing megabytes of external dependencies imposes prohibitive overhead.

To address these challenges, we introduce **NakedAgent**. NakedAgent explores a fundamental systems question: *Can an autonomous software engineering agent achieve robust, auditable, and extensible code modification loops while maintaining exactly zero third-party runtime dependencies?*

By confining its implementation exclusively to the Python standard library, NakedAgent guarantees zero supply-chain risk by construction. This paper details NakedAgent's architecture, its Mealy state machine execution lane, cryptographic Merkle event logging, offline replay verification, empirical results on SWE-bench Lite, and the *Omakase* extensibility doctrine.

2. SYSTEM ARCHITECTURE

2.1 Foundational Invariants

NakedAgent enforces two core invariants:

$$|D_{runtime}| = 0$$

stipulating that the agent requires zero external packages, and:

$$\forall t \in T_{foundation}: t: (A_{args}, W_{context}) \rightarrow R_{result}$$

where $T_{foundation}$ represents a minimal, closed set of four deterministic primitives.

2.2 The Four Foundation Tools

Rather than designing a sprawling surface of bespoke tools, NakedAgent isolates four primitives:

- `read [path]`: Inspects target files within the workspace boundary. Content is formatted with line numbers to anchor model attention, and directory traversal escapes are rejected fail-closed.
- `write [path]`: Writes or overwrites file contents atomically. Missing parent directories are created on demand.
- `patch [path]`: Executes targeted search-and-replace block edits. Unlike heuristic diff tools, `patch` enforces exact, unique substring matching:

$$\text{count}(S_{\text{search}}, F_{\text{content}}) = 1$$

If the search target occurs zero times (no match) or multiple times (ambiguity), the patch is rejected with an explicit error, preventing silent multi-site corruption.

- `shell [cmd]`: Executes system commands via `subprocess.run`. Interactive sessions enforce user confirmation before execution; non-interactive runs require an explicit `--allow-shell` flag and a prefix-based `--shell-allowlist`.

2.3 Code-Fence Tool Dispatch

Commercial agent frameworks rely heavily on provider-specific JSON function-calling schemas (e.g., OpenAI Tools, Anthropic Tool Use). However, localized open-weight models (such as Qwen 2.5-Coder 7B and Gemma 3 12B served via Ollama) often lack robust JSON schema adherence, resulting in serialization failures.

NakedAgent bypasses provider-native schema coupling through *Markdown Code-Fence Tool Parsing*. The agent instructs the model to invoke tools using standard triple-backtick fences tagged with the tool name:

```
```patch src/calculator.py
<<<<<<< SEARCH
def add(a, b):
 return a - b
=====
def add(a, b):
 return a + b
>>>>>>> REPLACE
```
```

A deterministic regular expression scanner extracts invocations from the generated stream. Because all instruct-tuned LLMs are pre-trained on code blocks, execution reliability is preserved across diverse model architectures without provider lock-in.

2.4 Transport Layer

Network I/O is managed entirely through `urllib.request`. NakedAgent natively supports local Ollama endpoints (default) and standard OpenAI-compatible API gateways (vLLM, LM Studio, hosted APIs) with zero third-party HTTP dependencies.

3. PROVABLE EXECUTION & REPLAY

3.1 The Mealy State Machine Reducer

To ensure determinism and formal tractability, NakedAgent models agent execution as a pure Mealy state machine:

$$\delta: (S_t, E_t) \rightarrow (S_{t+1}, [A_{t+1}])$$

where S_t is an immutable state record, E_t is an incoming event, and A_{t+1} is a sequence of outward actions. State transitions are purely functional and free of side effects; all environment I/O is isolated in a thin driver layer.

3.2 Merkle State Chaining

Every transition is logged sequentially to an append-only JSONL event log. Each event is cryptographically bound to the previous state using SHA-256 Merkle chaining:

$$H_0 = \text{SHA-256}(\text{GenesisHeader})$$

$$H_t = \text{SHA-256}(H_{t-1} \parallel \text{CanonicalJSON}(E_t) \parallel \text{StateDigest}(S_t))$$

This cryptographic linkage ensures tamper evidence: modifying any logged tool output or prompt retroactively invalidates all subsequent state hashes.

3.3 Zero-Inference Offline Replay

Recorded trajectories can be independently verified without invoking the LLM:

```
$ python -m nakedagent.replay run.jsonl
PASS run.jsonl: verified 5 events; final hash 6d1e09...
```

The replay engine executes the reducer δ against the recorded event stream, computing state hashes deterministically. If the final hash matches the log, the run is mathematically certified to have satisfied the operational semantics of the agent reducer.

Epistemic Scope: Cryptographic replay certifies **internal transcript causality and reducer determinism** ($\delta(S_t, E_t) \rightarrow S_{t+1}$).

It does not certify external semantic execution correctness or model immunity to prompt injection, which must remain defended via OS sandboxing.

4. THE OMAKASE DOCTRINE

NakedAgent adopts an *Omakase* ("chef's choice") design philosophy: providing curated, opinionated defaults that run out-of-the-box without decision fatigue, paired with a designed seam for customization.

4.1 The Plugin Seam

Extensibility requires no framework registration. Dropping a Python file into `.nakedagent/plugins/` (repo-local) or `~/.nakedagent/plugins/` (user-global) defining `TOOLS: dict` or `DISABLE: list` immediately modifies the agent's behavior. Workspace plugins are gated behind an explicit trust flag to protect the developer from malicious cloned repositories.

4.2 Dynamic Prompt Adaptation

Each tool function carries an intrinsic `.usage` attribute containing its Markdown example. When plugins are loaded at startup, the system prompt generator dynamically incorporates the active tool usage strings. Overriding a tool automatically updates its prompt documentation without manual prompt engineering. Furthermore, specifying `DISABLE = ['shell', 'write']` produces a provably read-only agent by pruning both dispatch and prompt surfaces.

5. SIDEKICK HARNESS & SAFETY GATES

For governed operational environments, NakedAgent implements:

- **Bounded Execution Budgets:** Fixed slot limits terminate trajectories predictably, preventing runaway loops.
- **Tri-State Escalation:** High-risk operations pass through a gate returning `ALLOW`, `DENY`, or `ESCALATE`. Escalation suspends execution until human review.
- **Zero-Dependency MCP Client:** A stdlib-only JSON-RPC 2.0 client supporting the Model Context Protocol over stdio.

6. EXPLORATORY PILOT: FAILURE MODES UNDER MINIMAL SCAFFOLDING

6.1 SWE-bench Lite Pilot Setup

We conducted an exploratory pilot on SWE-bench Lite [1] using an NVIDIA RTX 3080 Ti (12 GB VRAM), local Ollama serving `gemma3:12b`, and Astropy benchmark instances (`-12907`, `-14182`, `-14365`). We note that while mini-swe-agent ran at $T=0$, NakedAgent was evaluated under Ollama's default sampling temperature (≈ 0.8), representing an uncontrolled exploratory variable.

Table 1: SWE-bench Lite Exploratory Pilot Trajectory Results

| Configuration | 12907 | 14182 | 14365 | Patches | Resolved |
|-------------------------------------|---------------|----------------|---------------|-----------|----------|
| A1: mini-swe-agent (native tools) | BadRequest | BadRequest | BadRequest | 0/3 | 0/3 (0%) |
| A2: mini-swe-agent (text backticks) | FormatError | LimitsExceeded | FormatError | 0/3 | 0/3 (0%) |
| B: NakedAgent (bash parity) | Failed sed | Invalid regex | No edit | 0/3 | 0/3 (0%) |
| C1: NakedAgent (raw usage) | Bracket bug | Bracket bug | Bracket bug | 0/3 | 0/3 (0%) |
| C2: NakedAgent (CRLF pipe) | Patch applied | Patch applied | Patch applied | 3/3* | 0/3 (0%) |
| C3: NakedAgent (binary write) | 1-hunk patch | No patch | No patch | 1/3 valid | 0/3 (0%) |

*C2 generated whole-file CRLF rewrites due to host OS pipe conversion.

6.2 Failure Mode Taxonomy

The empirical pilot identified key failure modes distinguishing scaffold artifacts from model capability bounds:

1. **Provider-Native Rejection (A1):** Ollama local serving lacked JSON function-calling endpoints for Gemma 3, halting mini-swe-agent before execution began.
2. **Syntax Brittle Trapping (A2):** Demanding rigid bash formatting triggered format-recovery loops.
3. **Placeholder Literal Hallucination (C1):** When usage docs contained `<cmd>`, the model echoed angle brackets literally. Concrete usage examples eliminated this bug.
4. **Cross-OS Line Endings (C2):** Windows host pipes introduced CRLF characters into containerized Linux git repos, marking entire files as dirty. Enforcing binary stdin pipes restored clean diffs.
5. **Single-Hunk Patching (C3):** Binary I/O enabled Gemma 3 to localize and generate clean single-hunk diffs without external dependencies.
6. **Syntactic Compliance vs. Semantic Resolution:** While C3 produced a syntactically clean 1-hunk patch on `astropy-12907`, test-suite evaluation revealed 7 unit test regressions, resulting in 0/3 resolved tasks (0%). Minimal scaffolding successfully removes harness friction, but semantic problem-solving remains bounded by local model weights.

7. SECURITY ANALYSIS

Eliminating third-party dependencies eliminates dependency confusion, typosquatting, and transitive supply-chain poisoning. However, runtime defense requires strict containment:

- **Shell Hardening:** Commands undergo POSIX lexical splitting (`shlex.split`), defusing injection attacks via semicolons (`;`), pipelines (`|`), or substitutions (`${...}`). Automated runs enforce prefix allowlists.
- **Plugin Gating:** Workspace-local plugins in `.nakedagent/plugins/` are ignored by default and require explicit `--trust-plugins` opt-in, closing zero-click RCE vectors on untrusted repos.
- **Protected Surfaces:** The `write` and `patch` tools disallow modifying protected directories (`.git/`, `.nakedagent/`), and automated workloads require OS container sandboxing.

8. RELATED WORK

SWE-agent [2] demonstrated that specialized agent interfaces improve benchmark scores. Aider [3] pioneered terminal-based pair programming with git search/replace hunks. Krinea [9] formulated append-only invariants for verifiable governance receipts. NakedAgent bridges these domains, providing a zero-dependency implementation with cryptographic replay verification.

9. DISCUSSION & LIMITATIONS

Standard-library minimalism incurs conscious trade-offs: lack of asynchronous streaming UI rendering, absence of HTTP/2

multiplexing, and exact substring matching requirements on patches that require models to reproduce surrounding context accurately.

10. CONCLUSION & AVAILABILITY

NakedAgent proves that an autonomous coding agent can achieve full functionality, safety, and cryptographic auditability with zero third-party dependencies. Source code and replication artifacts are available at:

- GitHub: <https://github.com/hummb1-io/nakedagent>
- PyPI: `pip install nakedagent`
- Zenodo: [10.5281/zenodo.placeholder](https://zenodo.org/record/105281)

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